

Algorithms Problem Solving

Level 3

Convert Record to Line



Algorithms Problem Solving

Level 3

Convert Record to Line

Definition

Struct Serialization

 Program converts an `sClient` structure into one delimited text line.

 This process is called serialization because structured data is transformed into a storable text representation.

Explanation

Struct Field Access

-  The function reads fields such as ``AccountNumber``, ``PinCode``, ``Name``, ``Phone``, and ``AccountBalance``.

-  Each field is prepared for insertion into the output line.

Type Conversion

-  Numeric values such as ``AccountBalance`` must be converted into strings.

-  ``to_string()`` can perform this conversion automatically.

Delimiter-Based Formatting

-  Fields are concatenated using a separator such as ``"#//#"``.

-  The delimiter allows the record to be separated again later.

File Storage Preparation

-  The resulting string can be written directly as one line in a text file.

-  The same delimiter must be used when deserializing the line.

Procedures

1. Define the `sClient` structure.
2. Create or receive a client record.
3. Read each struct field.
4. Convert numeric fields to strings.
5. Concatenate fields using `#/#`.
6. Return the resulting line.

💡 Example

Client Record:

Acc: A150

PIN: 1234

Name: Ahmad

Phone: 079

Balance: 5000

➡ Result

A150#/#1234#/#Ahmad#/#079#/#5000.000000

Note

- ◆ Serialization allows structured memory data to be stored in a simple text file.
- ◆ `to_string(double)` may produce trailing zeros such as `5000.000000`.
- ◆ The chosen delimiter becomes part of the file format and must remain consistent.

⊘ Warning

- ◆ The delimiter should not naturally appear inside data fields unless escaping or another encoding strategy is implemented.
- ◆ Incorrect field ordering can cause the deserialization function to reconstruct the wrong values.
- ◆ Numeric formatting may need custom handling if a specific number of decimal places is required.



Comparison

▣ `ConvertClientDataToLine()`

- ✓ Converts a struct into a delimited string.
- ✓ Performs serialization.
- ✓ Prepares data for text-file storage.

▤ `ConvertLineToRecord()`

- ✓ Converts a delimited string back into a struct.
- ✓ Performs deserialization.
- ✓ Restores structured data from stored text.

✦✦ Summary

- ✓ Reads structured client fields.
- ✓ Converts non-string fields into text.
- ✓ Joins all fields using a consistent delimiter.
- ✓ Produces a single serialized line suitable for file storage.

Thank you 😊